

KARTHIK M

karthik.nit1986@gmail.com · (469) 617-8954 · linkedin.com/in/karthikmari

Summary

Data Engineer with 11+ years of experience designing and delivering scalable, end-to-end data pipelines across cloud and on-premise platforms. Deep expertise in Python, PySpark, SQL, ETL/ELT development, and multi-source data ingestion (relational databases, Spark, S3, XML, JSON). Proven track record of orchestrating automated pipelines with Airflow, applying data validation and quality frameworks, and collaborating with cross-functional stakeholders to translate complex data problems into business solutions. Committed to reproducible, well-documented code artifacts and modern software engineering practices.

Work Experience

Senior Data Engineer — Wells Fargo (Jun 2024–Present)

- Built scalable PySpark ETL pipelines to ingest, cleanse, and validate 5+ TB of customer and operational data, aligning to modern software development principles for downstream analytical consumption.
- Orchestrated continuous data delivery using Apache Airflow, Azure Data Factory, and Databricks Workflows, reducing compliance report generation time by ~40%.
- Established data lineage and metadata management through Unity Catalog governance, enabling reliable traceability across all data assets.
- Designed standardized data schemas in Hackolade and harmonized multi-vendor feeds to ensure data quality and suitability for analytical use cases.
- Rewrote legacy SQL into optimized Spark and dbt transformations, improving daily pipeline execution by ~35%.
- Implemented CI/CD with GitHub and Azure DevOps to produce reproducible code artifacts and streamline hand-off to downstream data science teams.

Notable work: Led the Lakehouse migration and governance rollout that standardized access controls and automated lineage for compliance reporting.

Senior Data Engineer — Medtronic (Nov 2021–May 2024)

- Architected real-time data pipelines for embedded pacemaker sensor data using Apache Kafka for buffering and PySpark Streaming for processing, enabling ingestion into front-end analytical systems.
- Built REST APIs exposing aggregated telemetry for patient-facing dashboards, translating complex technical data into consumable business solutions.
- Integrated streaming and batch pipelines in Azure Synapse and Databricks to deliver synchronized, high-quality data assets for analytics and APIs.

- Automated file movement, splitting, and validation with Python to enforce data quality across ingestion workflows.

Azure Data Engineer — State Farm (Apr 2020–Oct 2021)

- Developed scalable data pipelines to ingest from relational databases (DB2, SQL Server, Oracle), flat files, REST APIs, XML, and JSON — covering the full spectrum of structured and semi-structured data sources.
- Implemented real-time streaming ingestion with Azure EventHub and Spark Streaming for time-sensitive analytical use cases.
- Applied exploratory data analysis to evaluate nulls, duplicates, and data quality issues prior to pipeline promotion.
- Provisioned infrastructure with Terraform and implemented CI/CD using Azure DevOps YAML pipelines, supporting reproducible deployments.

Data Engineer — Centene Corporation (Jul 2018–Mar 2020)

- Built and operationalized large-scale ingestion, transformation, validation, and orchestration workflows for real-time processing applications.
- Partnered with DevSecOps and cross-functional IT stakeholders (database, operations, technical support) to standardize staging, provisioning, and decommissioning procedures.
- Contributed to Agile delivery by enabling CI/CD pipelines and maintaining continuous data delivery for analytics consumers.

Senior ETL Developer — Nielsen (Oct 2016–Jun 2018)

- Architected serverless ETL processing handling ~55 TB/day using Apache Spark and AWS S3, contributing to a multi-petabyte data warehouse.
- Identified and validated data sources for quality and availability at enterprise scale, ensuring suitability for analytical use cases.

Data Warehouse Developer — WellCare Health Plans (Jan 2015–Sep 2016)

- Developed and maintained data warehouse solutions and ETL pipelines to support healthcare reporting and analytics.

Data Warehouse Developer — Truist (Jan 2014–Dec 2014)

- Built and supported data warehouse ETL components for banking data integration and reporting.
-

Skills

- **Languages & frameworks:** Python, PySpark, SQL, T-SQL, C#, Java
- **Data ingestion & ETL:** Multi-source ingestion (relational DBs, Spark, S3, XML, JSON), Delta Lake, Azure Data Factory, Azure Synapse, Snowflake, Kafka
- **Metadata & data quality:** Unity Catalog (lineage & metadata management), Great Expectations, dbt, Delta Live Tables

- **Orchestration & automation:** Apache Airflow, Databricks Workflows, Azure Data Factory
 - **DevOps & version control:** GitHub, Azure DevOps, Terraform, Docker, CI/CD, YAML pipelines
 - **Cloud platforms:** Azure (Data Lake, Synapse, ADF, EventHub), AWS (S3)
 - **BI & reporting:** Power BI, Tableau
-

Education

Master of Science, Computer Science — Stevens Institute of Technology (Dec 2013)

Certifications

- Databricks Certified Data Engineer (Associate & Professional)